Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\,\boldsymbol{\delta x[k]} - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^{d} \delta x_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Big\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$
$$- \underbrace{\Big[\delta x_D[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \delta x_D[k-i]\Big]}_{\delta x_D^d[k]} \Big\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\Big\{a_x[k] f_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i]\Big\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big] - \delta x_D^d[k]$$

$$a_x[k]\,f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = e_{a_x}[k]\,f_{dx}[k] + (1-\lambda_c) n_x[k] \\ + (1-\lambda_c)\textstyle\sum_{i=1}^{d}\big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \qquad a_{x_{ram}}[k] \approx a_x'[k]\,x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= \boldsymbol{a_{x_{det}}[k+1]} + a_{x_{ram}}[k+1] \\ &= \underbrace{\Big\{\boldsymbol{a_{x_{det}}[k]} + a_{x_{ram}}[k]\Big\}}_{a_x[k]} + \boldsymbol{\Delta a_x[k]} + \underbrace{\Big\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\Big\}}_{\Delta a_{ram}[k]} \end{aligned}$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a_x'[k]\,\Delta h_d[k], \qquad \Delta a_{ram}[k] = a_x'[k]\,\Delta x_{ram}[k]$$

$\Delta h_d[k] = h_d[k+1] - h_d[k]$ (desired-trajectory height increment).

Parameter estimator: $\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k]$

Parameter estimation error: $e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$

Estimation error dynamics: $e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k]$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^{i} \Delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k]\, & \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ & - \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i)\, f_{dx}[k-i]\, \Delta a_{ram}[k-i] \right\}}_{\delta x^d_{\delta af}[k]} \\ & + \delta x^d_D[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x^d_D[k] - \underbrace{\left\{ \delta x^d_T[k] - \delta x^d_{\delta af}[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]}}$$

State equation

($\delta x^d_D \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^{d} \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^{d} \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^{d} \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \qquad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \Big(\hat{a}_x[k] - \textstyle\sum_{i=1}^{d} \Delta a_x[k-i]\Big) \\ &= \Big(a_x[k] - \textstyle\sum_{i=1}^{d} \Delta a_x[k-i] + r_2[k]\Big) - \hat{a}_x[k] + \textstyle\sum_{i=1}^{d} \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\Big\{n_a[k] - \textstyle\sum_{i=1}^{d} \Delta a_{ram}[k-i]\Big\}}_{r_2[k]} \end{aligned}$$

$$\delta\hat{x}_1[k+1] = \delta\hat{x}_2[k] + \ell_{11}[k]\, e_{y_1}[k] + \ell_{12}[k]\, e_{y_2}[k]$$

$$\delta\hat{x}_2[k+1] = \delta\hat{x}_3[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k]$$

$$\delta\hat{x}_3[k+1] = \lambda_c \cdot \delta\hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k]\, e_{\delta x_1}[k] - \ell_{12}[k]\, e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k]\, e_{\delta x_1}[k] - \ell_{22}[k]\, e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k]\, e_{\delta x_1}[k] - \ell_{32}[k]\, e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = e_{a_x}[k] - \ell_{41}[k]\, e_{\delta x_1}[k] - \ell_{42}[k]\, e_{a_x}[k] + \underbrace{\Delta a_{ram}[k]}_{q_4[k]}$$

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^{d} f_{dx}[k-i]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k]\,\mathbf{r}[k]$$

Gain fluctuation variance $\mathrm{Var}(\Delta a_{ram})$

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$.

$$a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}, \qquad K_h = \frac{1}{c}\frac{dc}{d\bar{h}} \text{ (wall sensitivity)}, \qquad a_{x_{ram}} = a'_x\, x_{ram}, \quad x_{ram} = -\delta x$$

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\mathrm{Var}(x_{ram}) = \mathrm{Var}(\delta x) = C_{\delta x}\,\sigma^2_{\delta h}, \qquad C_{\delta x} = 2 + \frac{1}{1-\lambda_c^2}, \qquad \sigma^2_{\delta h} := 4k_B T\, a_x$$

Increment of a stationary signal ($\rho_1 = \mathrm{Cov}(x_{ram}[k{+}1], x_{ram}[k])/\mathrm{Var}(x_{ram})$, lag-1 autocorr):

$$\mathrm{Var}(\Delta x_{ram}) = 2\,\mathrm{Var}(x_{ram}) - 2\,\mathrm{Cov}\big(x_{ram}[k{+}1], x_{ram}[k]\big) = 2(1-\rho_1)\,\mathrm{Var}(x_{ram})$$

$$\rho_1 = \lambda_c + \frac{2(1-\lambda_c)}{C_{\delta x}} \;\Rightarrow\; 1-\rho_1 = \frac{1}{(1+\lambda_c)\,C_{\delta x}}$$

Gain increment $\Delta a_{ram} = a'_x\,\Delta x_{ram}$ ($C_{\delta x}$ and $(1-\rho_1)$ cancel):

$$\mathrm{Var}(\Delta a_{ram}) = a'^2_x\,\mathrm{Var}(\Delta x_{ram}) = a'^2_x \cdot \underbrace{2(1-\rho_1)\,C_{\delta x}}_{=2/(1+\lambda_c)}\,\sigma^2_{\delta h}$$

$$\boxed{\mathrm{Var}(\Delta a_{ram}) = \frac{2}{1+\lambda_c}\,a'^2_x\,\sigma^2_{\delta h}}$$

Impl.: $K_h,\ \sigma^2_{\delta h} = 4k_B T\,a_\perp$ evaluated at $\bar{h}_{\mathrm{meas}}$ (measured, p_m); gain $a_x \to \hat{a}_x$ (estimate).

Correction — the R_{22} delay term telescopes

$$\Delta a_{ram}[k{-}1] + \Delta a_{ram}[k{-}2] = a_{ram}[k] - a_{ram}[k{-}2] \;\Rightarrow\; \mathrm{Var}\Big(\textstyle\sum_{i=1}^{2}\Delta a_{ram}[k{-}i]\Big) = 2\,(1{-}\rho_2)\,\mathrm{Var}(a_{ram})$$

$$\boxed{R_{22}^{delay} = \frac{1-\rho_2}{2\,(1-\rho_1)}\sum_{i=1}^{d}\mathrm{Var}\big(\Delta a_{ram}[k-i]\big), \qquad \frac{1-\rho_2}{2(1-\rho_1)} = 1.105\ (\lambda_c{=}0.7)}$$

$\rho_1 = 0.8515, \quad \rho_2 = 0.6718 \qquad$ (ρ_τ = lag-τ autocorr of δx; independent sum $\Rightarrow -9.5\%$)

Impl.: offline constant `r22_delay_sum_factor` (build_eq17_6state_constants); $d = 1 \to 1$ (single term, exact).

Correction — $a_{x_{ram}}$ is first-order autoregressive (AR(1)), not a random walk

True-signal dynamics ($x_{ram} = -\delta x$; $\delta x[k+1] = \lambda_c\, \delta x[k] - F_{dx}[k]\, e_{a_x}[k] - \varepsilon[k]$):

$$a_{x_{ram}}[k+1] = -a'_x\, \delta x[k+1] = \lambda_c\, a_{x_{ram}}[k] + a'_x\, F_{dx}[k]\, e_{a_x}[k] + a'_x\, \varepsilon[k]$$

$$\Delta a_{ram}[k] = (\lambda_c - 1)\, a_{x_{ram}}[k] + a'_x\, F_{dx}[k]\, e_{a_x}[k] + a'_x\, \varepsilon[k]$$

True gain state ($a_x = a_{x_{det}} + a_{x_{ram}}$, $a_{x_{det}}[k+1] = a_{x_{det}}[k] + \Delta a_x[k]$, $e_{a_x} = a_x - \hat{a}_x$):

$$\begin{aligned} a_x[k+1] &= \big(a_{x_{det}}[k] + \Delta a_x[k]\big) + \lambda_c\big(a_x[k] - a_{x_{det}}[k]\big) + a'_x F_{dx}[k]\big(a_x[k] - \hat{a}_x[k]\big) + a'_x\, \varepsilon[k] \\ &= \underbrace{\big(\lambda_c + a'_x F_{dx}[k]\big)\, a_x[k]}_{\mathbf{F}_e(4,4)} + \underbrace{(1-\lambda_c)\, a_{x_{det}}[k] + \Delta a_x[k] - a'_x F_{dx}[k]\, \hat{a}_x[k]}_{\text{known input}} + \underbrace{a'_x\, \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e(4,4) = \lambda_c + a'_x F_{dx}[k]}$$

Estimator ($\big(\lambda_c + a'_x F_{dx}\big)\hat{a}_x - a'_x F_{dx}\, \hat{a}_x = \lambda_c\, \hat{a}_x$: the feedthrough vanishes at the estimate):

$$\hat{a}_x[k+1] = \lambda_c\, \hat{a}_x[k] + (1-\lambda_c)\, a_{x_{det}}[k] + \Delta a_x[k] + \ell_{4\cdot}[k]\, e_y[k]$$

$$e_{a_x}[k+1] = a_x[k+1] - \hat{a}_x[k+1] = \big(\lambda_c + a'_x F_{dx}[k]\big)\, e_{a_x}[k] + a'_x\, \varepsilon[k] - \ell_{4\cdot}[k]\, e_y[k]$$

Process noise (white driver $w_4 = a'_x \varepsilon$; equivalent-white $\mathrm{Var}_{\mathrm{eff}}(\varepsilon) = (1-\lambda_c^2)\, \mathrm{Var}(\delta x)$, stationary-variance matching — the per-step $\gamma_\varepsilon(0) = (1 + 2(1-\lambda_c)^2)\, \sigma^2_{\delta h}$ differs by 1.712; ε is MA(d), δx is not AR(1)):

$$q_4 = \mathrm{Var}(w_4) = a'^2_x\, \mathrm{Var}(\varepsilon) = a'^2_x\, (1-\lambda_c^2)\, \mathrm{Var}(\delta x) = (1-\lambda_c^2)\, C_{\delta x}\, a'^2_x\, \sigma^2_{\delta h}$$

$$\boxed{q_4 = a'^2_x\, \mathrm{Var}(\varepsilon) = (3 - 2\lambda_c^2)\, a'^2_x\, \sigma^2_{\delta h}}$$

Impl.: predict and q_4 use $\hat{a}_x$ with a'_x at $\bar{h}_{\mathrm{meas}}$ (p_m); reversion target $a_{x_{det}}$ from p_d; production $\mathbf{F}_e(4,4) = \lambda_c$ (drops $a'_x F_{dx}$).

Taylor gain — the full first-order model (anchor-free)

Gain by Taylor expansion about the desired trajectory ($x = x_d - \delta x$):

$$a_x[k{+}1] = a_x\big(x[k{+}1]\big) = \underbrace{a_x\big(x_d[k+1]\big)}_{a_{x_{det}}[k+1]} - a'_x[k{+}1]\,\delta x[k{+}1], \qquad a_{x_{det}}[k{+}1] = a_{x_{det}}[k]{+}a'_x[k]\,\Delta h_d[k], \qquad a'_x[k{+}1] \approx$$

True gain state ($a_{x_{det}}[k] = a_x[k] + a'_x\,\delta x[k]$, then the boxed δx recursion):

$$\boxed{a_x[k+1] = a_x[k] + a'_x\,\Delta h_d[k] + (1-\lambda_c)\,a'_x\,\delta x[k] + a'_x\,F_{dx}[k]\,e_{a_x}[k] + a'_x\,\varepsilon[k]}$$

Estimator ($E\{e_{a_x}\} = 0$, $E\{\varepsilon\} = 0$; $\delta x[k] \to \delta\hat{x}_3[k]$):

$$\boxed{\hat{a}_x[k+1] = \hat{a}_x[k] + a'_x\,\Delta h_d[k] + (1-\lambda_c)\,a'_x\,\delta\hat{x}_3[k] + \ell_{41}[k]\,e_{y_1}[k] + \ell_{42}[k]\,e_{y_2}[k]}$$

Error dynamics ($e_{\delta x_3} = \delta x_3 - \delta\hat{x}_3$):

$$\boxed{e_{a_x}[k+1] = \big(1 + a'_x F_{dx}[k]\big)\,e_{a_x}[k] + (1-\lambda_c)\,a'_x\,e_{\delta x_3}[k] + \underbrace{a'_x\,\varepsilon[k]}_{q_4[k]} - \ell_{41}[k]\,e_{y_1}[k] - \ell_{42}[k]\,e_{y_2}[k]}$$

$\mathbf{F}_e$ **and process noise** ($q_3 = -\varepsilon$, $q_4 = a'_x\,\varepsilon$: one sample $\Rightarrow$ rank-1):

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & (1-\lambda_c)\,a'_x & 1 + a'_x F_{dx}[k] \end{bmatrix}, \qquad \mathrm{eig}(\mathbf{F}_e) = \{\,0,\ 0,\ \lambda_c + a'_x F_{dx},\ 1\,\}}$$

$$\boxed{Q_{33} = \sigma_\varepsilon^2, \qquad Q_{34} = Q_{43} = -\,a'_x\,\sigma_\varepsilon^2, \qquad Q_{44} = a'^2_x\,\sigma_\varepsilon^2, \qquad \sigma_\varepsilon^2 := \gamma_\varepsilon(0) = \big(1 + 2(1-\lambda_c)^2\big)\sigma_{\delta h}^2}$$

$P_{44} = a'^2_x\,P_{33}, \qquad P_{34} = -a'_x\,P_{33} \qquad$ (exact filter invariant, $\forall R_{22}$)

$\frac{\gamma_\varepsilon(0)}{1-\lambda_c^2} = 2.31\,\sigma_{\delta h}^2$ vs $\mathrm{Var}(\delta x) = C_{\delta x}\,\sigma_{\delta h}^2 = 3.96\,\sigma_{\delta h}^2$ (ε colored: white-ε model under-represents $\mathrm{Var}(a_{ram})$)

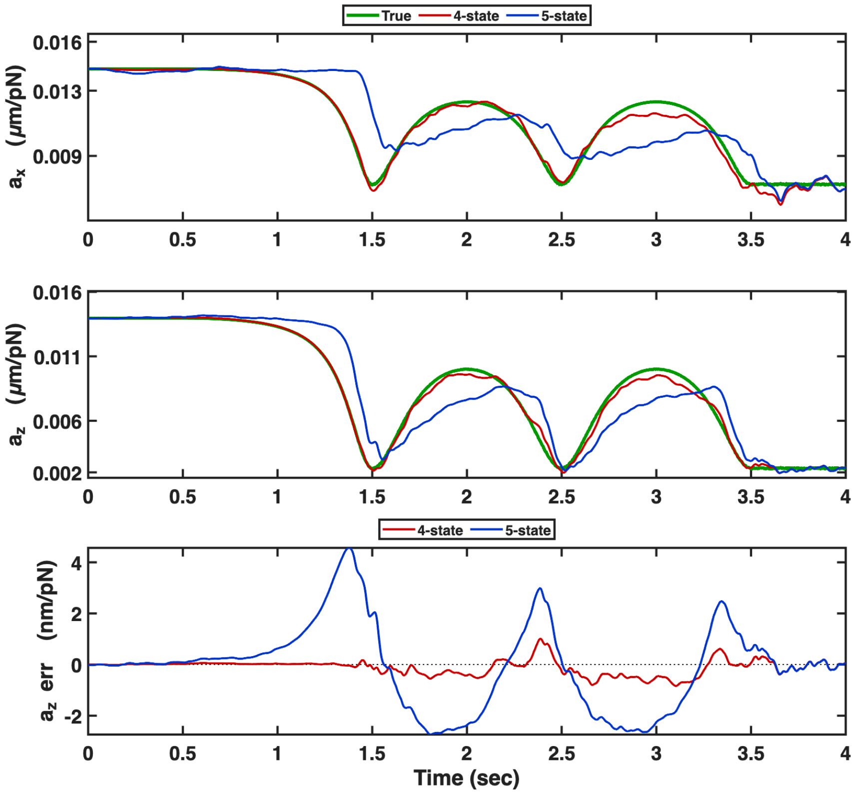
Feedforward (implemented forms; raw $a'\Delta h$ accumulates ~3.6% Euler drift over the descent):

known: $\Delta\hat{a} = a_{x_{det}}[k] - a_{x_{det}}[k-1]$; $\qquad$ ahat: $\Delta\hat{a} = \hat{a}_x\Big(\frac{a_{x_{det}}[k]}{a_{x_{det}}[k-1]} - 1\Big)$; $\qquad$ diff: $\Delta\hat{a} = \hat{a}'_x\,\Delta h_d[k-1]$

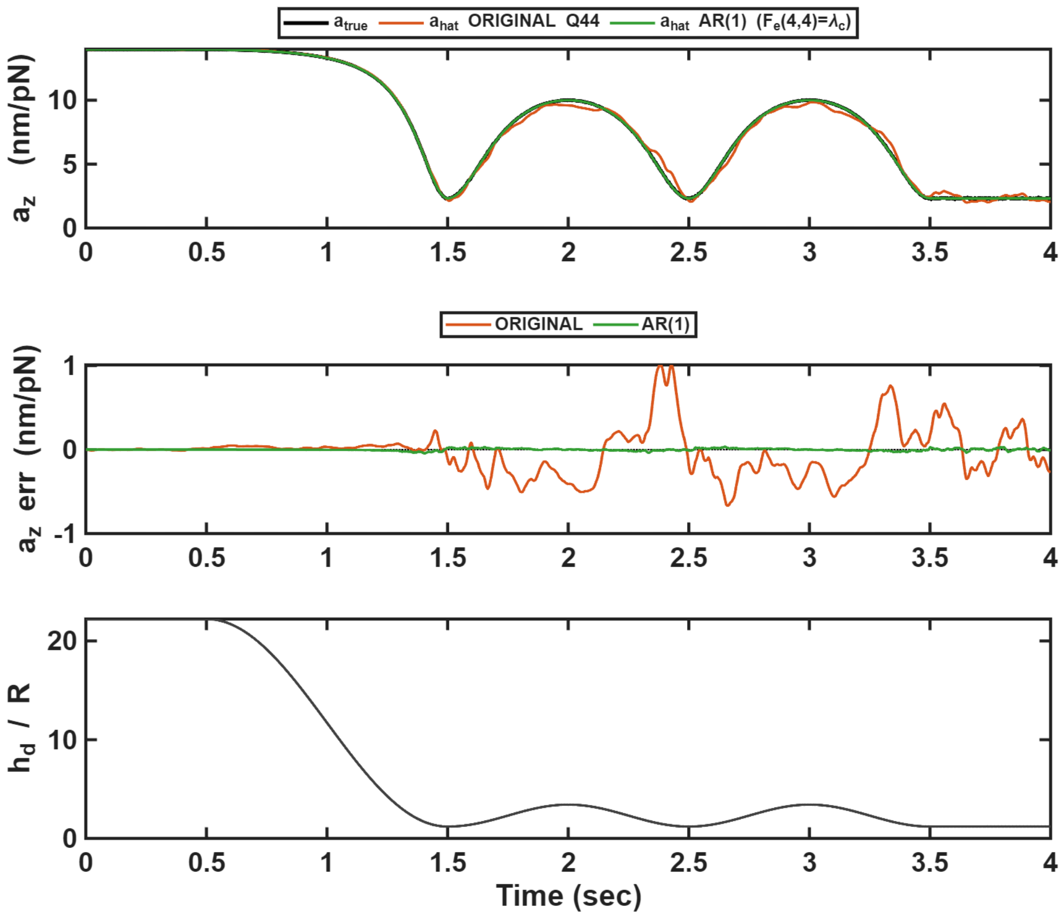
Impl.: knobs `use_taylor_gain` + `aprime_source` (known | ahat | diff = the self-difference box below), default off. z axis: full row 4. x, y: the gain argument is the wall-normal height, so z's $\delta\hat{x}_3$ enters their predict as a known input; row 4 = $[\,0\ 0\ 0\ 1\,]$, $Q_{34} = 0$, $Q_{44} = a'^2_i\,\sigma_{\varepsilon,z}^2$.

Verification (1 Hz, near wall)

Per-axis EKF, seed ensemble; near-wall 1 Hz oscillation ($h_{init} = 50\,\mu$m, trough $\bar{h} = 1.2$).



a_x, a_z: true gain and the 4-state / 5-state estimates; bottom tile, the a_z estimation error $\hat{a}_x - a_x$.



a_z: original random-walk Q_{44} ($\mathbf{F}_e(4,4) = 1$) vs AR(1) reverting gain ($\mathbf{F}_e(4,4) = \lambda_c$); middle tile, estimation error; bottom, $\bar{h}$. Near-wall $|\hat{a}_z - a_z|$: 9.4%→0.4%.

Model-free a' via self-difference — original (RW) vs AR(1)

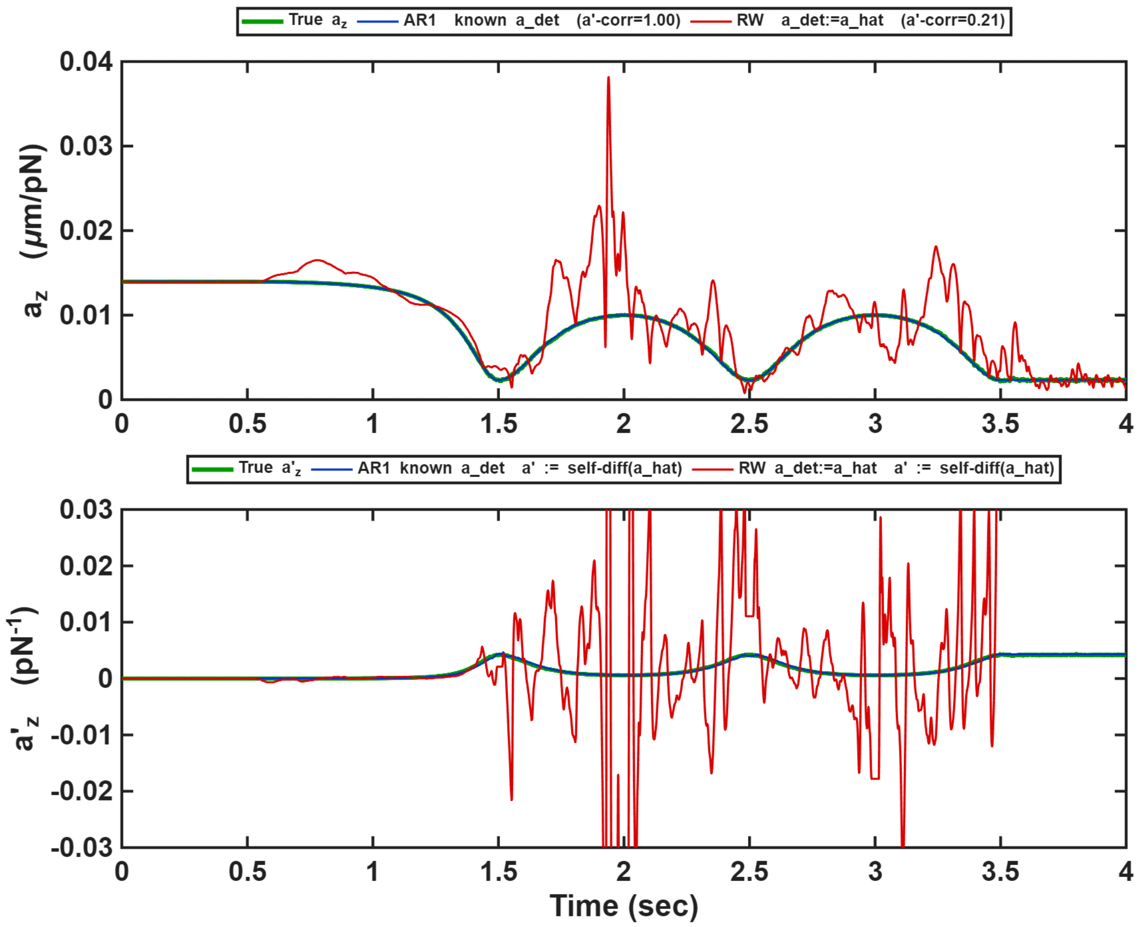
So far a'_x has come from the known wall model. If the wall is unknown, we can read a'_x straight off the gain estimate instead: take the change in $\hat{a}_x$ and divide it by the (known) change in desired height. But that raw ratio is very jumpy — $\hat{a}_x$ already carries estimation noise, and taking a difference makes the noise worse. So we do two simple things: (i) smooth the raw slope with a running average (an EWMA with weight β), and (ii) stop updating whenever the height is barely moving (a gate γ, so we never divide by almost zero):

$$\boxed{\hat{a}'_x[k] = \begin{cases} (1-\beta)\,\hat{a}'_x[k{-}1] \;+\; \beta\,\dfrac{\hat{a}_x[k]-\hat{a}_x[k{-}1]}{\Delta h_d[k{-}1]}, & |\Delta h_d[k{-}1]| > \gamma \\[2ex] \hat{a}'_x[k{-}1], & |\Delta h_d[k{-}1]| \le \gamma \quad \text{(gate closed: keep the last value)} \end{cases}}$$

$$\beta = 0.05 \quad \text{(running-average weight: smaller = smoother but slower)} \qquad \gamma = 10^{-3}\,\mu\text{m} \quad \text{(freeze } \hat{a}' \text{ once } |\Delta h_d| \text{ drops below this)}$$

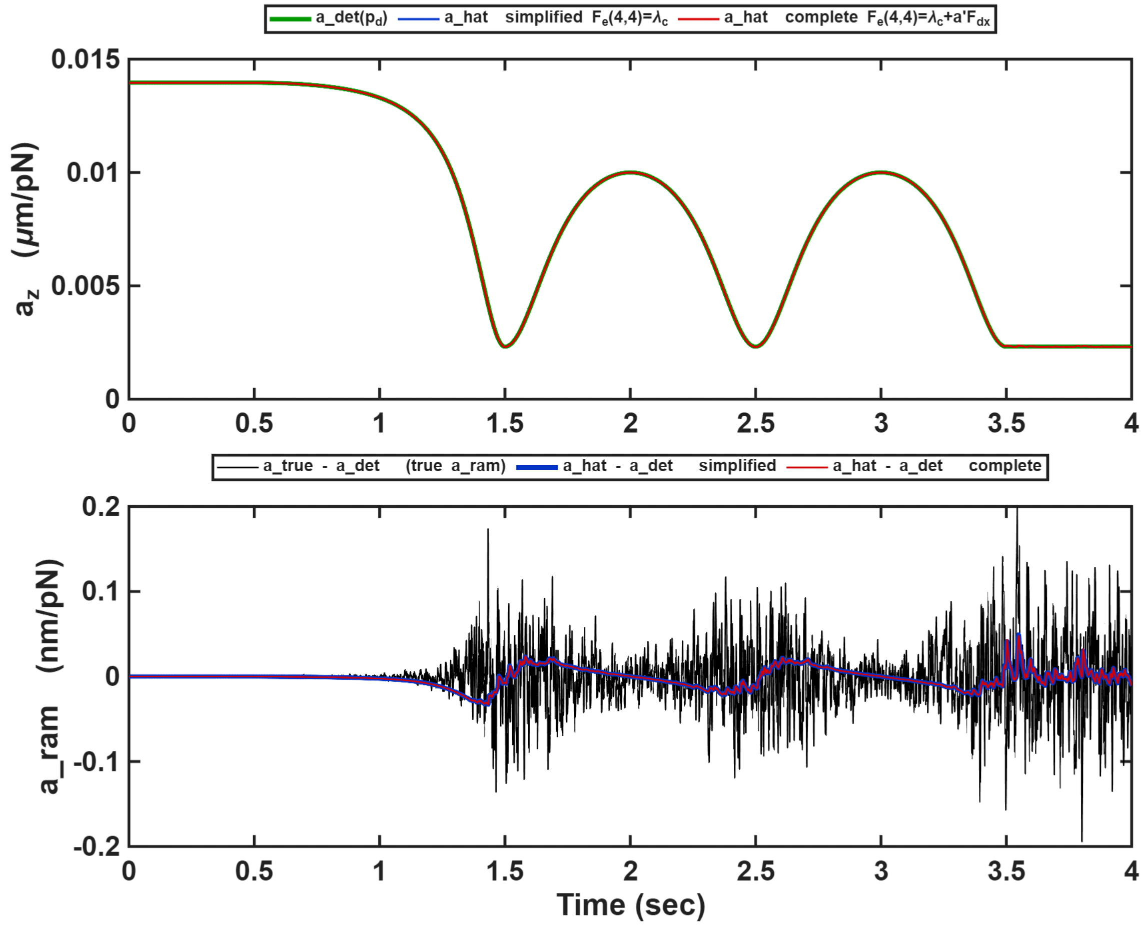
This one self-difference supplies a' everywhere in both runs; the runs differ *only* in how the gain *level* $a_{x_{det}}$ is anchored:

- **original (RW)**: $a_{x_{det}} := \hat{a}_x$ (no outside anchor — the estimate leans on itself), $\mathbf{F}_e(4,4) = 1$, $Q_{44} = \dfrac{2}{1+\lambda_c}\,\hat{a}'^2_x\,\sigma^2_{\delta h}$ — *fully* model-free.
- **AR(1)**: the level $a_{x_{det}} = a_{\text{nom}}/c(\bar{h}_d)$ is still taken from the wall, $\mathbf{F}_e(4,4) = \lambda_c$, $Q_{44} = (3 - 2\lambda_c^2)\,\hat{a}'^2_x\,\sigma^2_{\delta h}$ — a' is model-free, but the level still needs the wall.



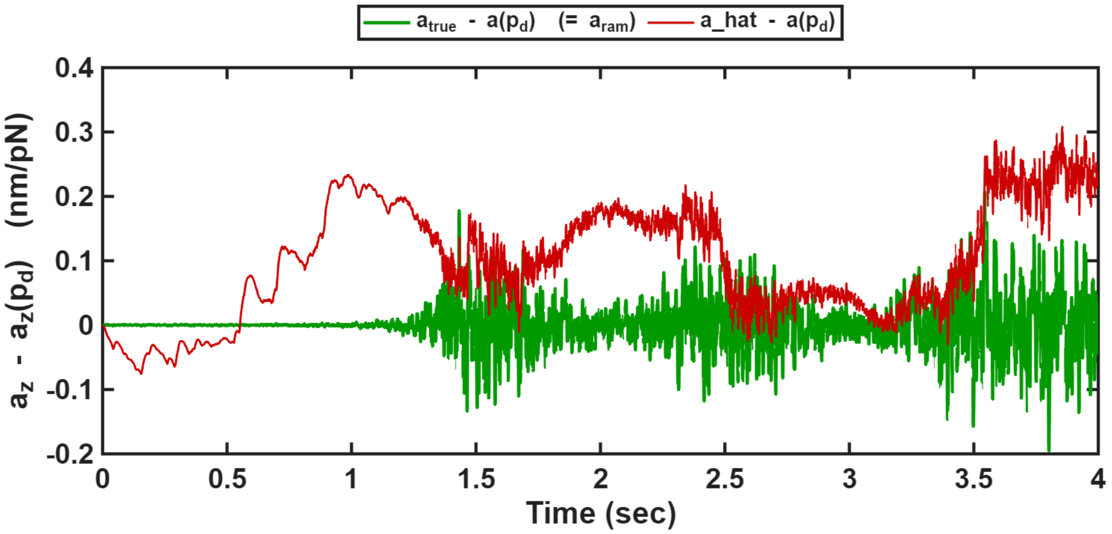
Same self-difference $\hat{a}'_x$ in both runs. Top: gain a_z. Bottom: slope a'_z. With a known level anchor (AR(1)) the estimate follows the truth; without one (RW) it is buried in noise.

Verification — $\mathbf{F}_e(4,4) = \lambda_c$ (simplified) vs $\lambda_c + a'_x F_{dx}$ (complete), known $c(\bar{h})$, same seed

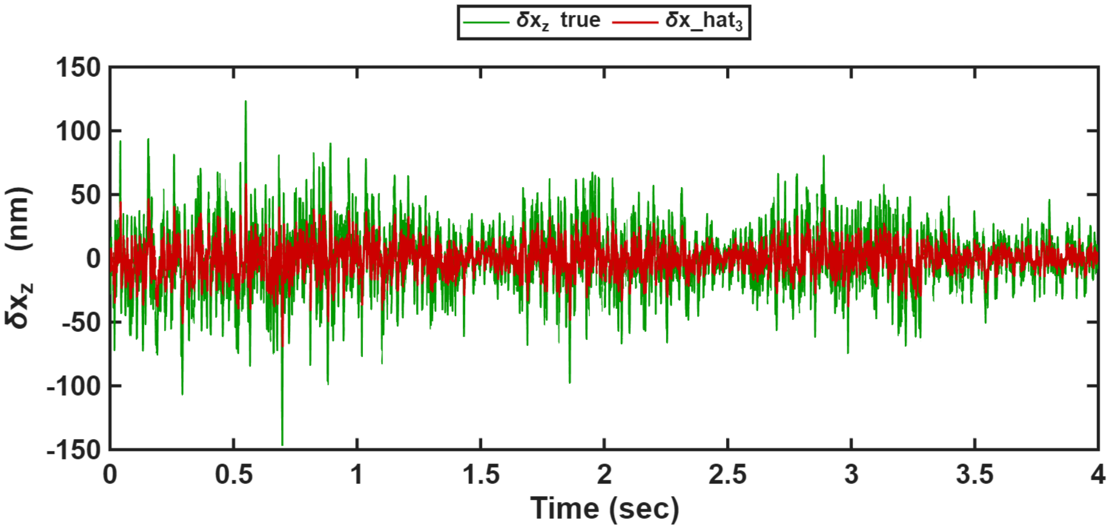


Top: a_z *—* $a_{x_{det}}(p_d)$ *and the two* $\hat{a}_x$*. Bottom:* $a_x - a_{x_{det}}$ *(true, black) and* $\hat{a}_x - a_{x_{det}}$ *(both versions). The two versions coincide:* $\max|\hat{a}_x^{\text{complete}} - \hat{a}_x^{\text{simplified}}|/a_{x_{det}} = 0.085\,\%$*;* $a_{x_{det}}(p_d) - \hat{a}_x = -0.03\,\% \pm 0.30\,\%$ *(*$\bar{h}_d < 2$*:* $\pm\,0.41\,\%$*).*

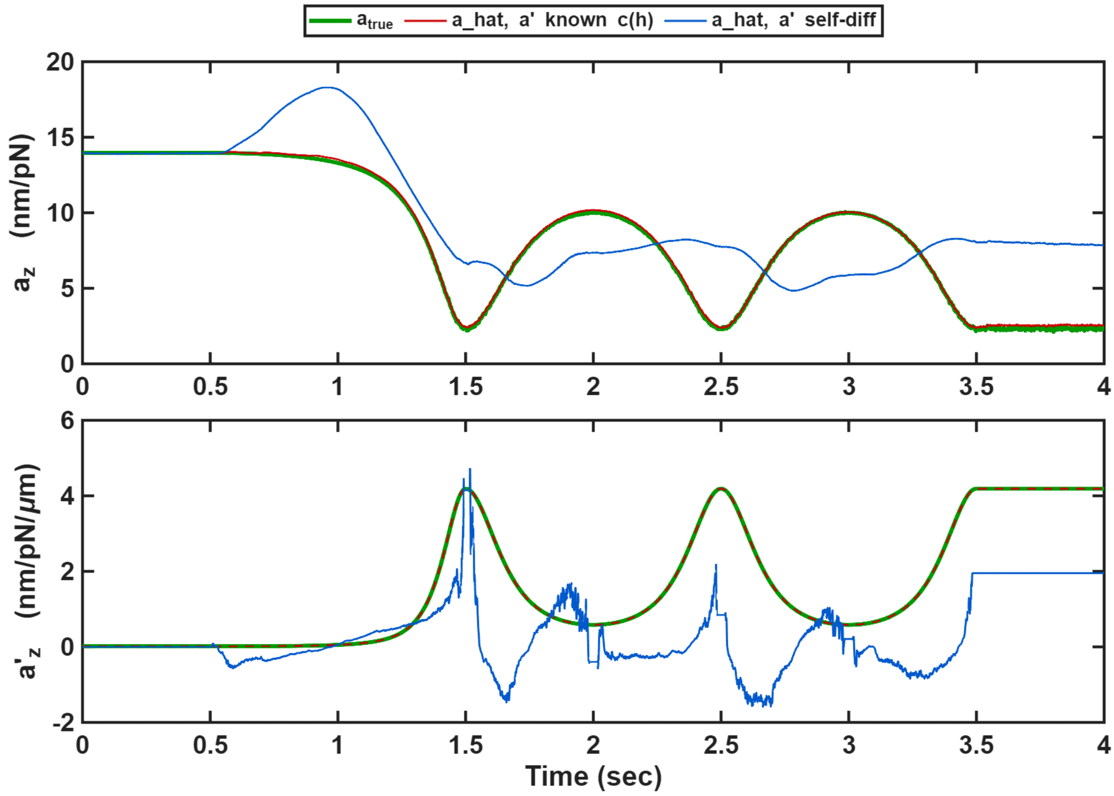
Verification — taylor gain (z axis, 1 Hz osc, seed 1)



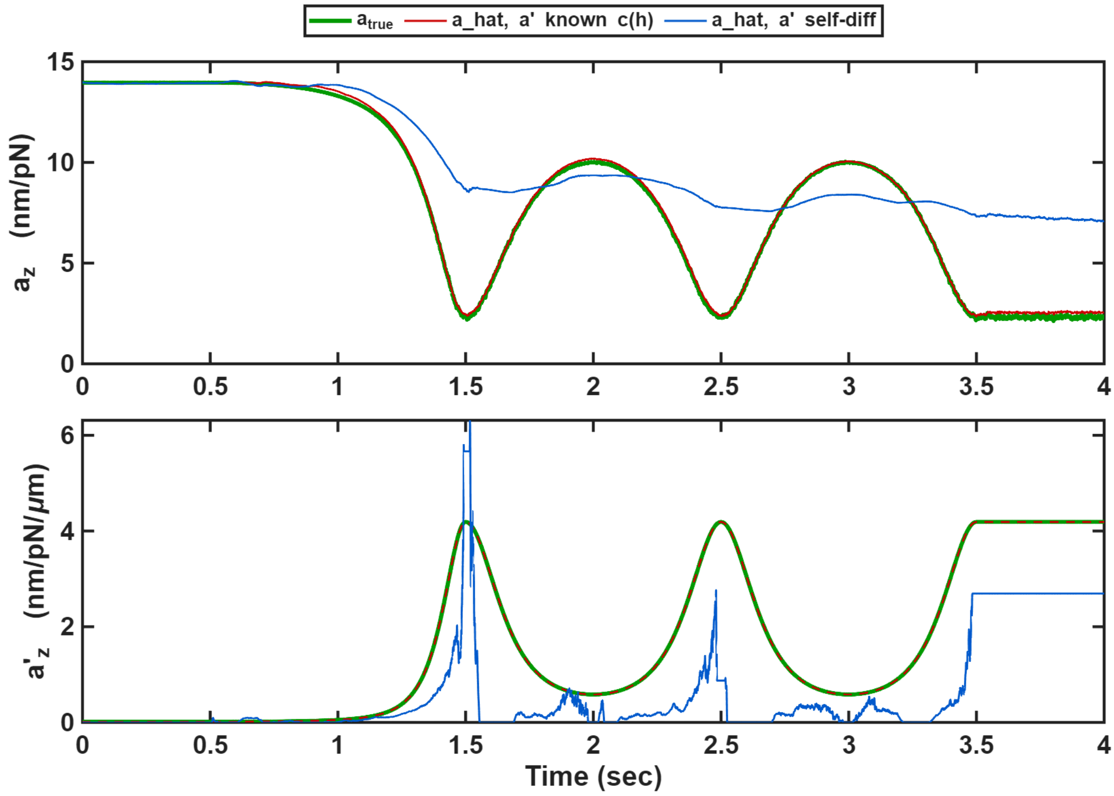
Oracle-a′ arm: $\hat{a}_x - a_x(p_d)$ vs $a_{x,true} - a_x(p_d)$ $(= a_{x_{ram}})$.



Same arm: $\delta\hat{x}_3$ vs true δx_z.



$\hat{a}_x$ *(top) and* a'_x *(bottom): oracle-*a' *arm vs self-diff arm (no clamp).*



Same seed with the post-EWMA clamp $\hat{a}'_x \leftarrow \max(\hat{a}'_x, 0)$*. Six-arm study (20 seeds, osc window): taylor-known* $-0.69\% \,/\, 3.08\%$ *vs RW* $-2.5\% \,/\, 16.5\%$ *vs AR(1)* $+0.02\% \,/\, 0.96\%$ *(bias/std); self-diff* $+33\% \,/\, 90\%$*.*